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Launcher Tipoff Deflection

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Introduction

The design of a rocket launcher rail is constrained by several considerations. First, the rail must be restrained to preclude any rigid body rotations when the rocket launches. Second, while the rail will deflect during launch, such deflections shall be required to not exceed a specified value. The allowable upper bound for rail tip deflection during launch has been chosen to be 0.1^{o}. This memo addresses the second point.

To keep the analysis brief, we assume the beam deflects statically even though the time scale for the rocket to accelerate to the end of the rail is of the same order as the rail's fundamental bending period. The maximum bending deflection will occur just before the rocket leaves the rail. This corresponds to the maximum bending moment applied to the greatest length of beam. The desired result is the slope at the free end of the rail at the instant when the rocket separates. This angle is, in effect, a change in Quadrant Elevation angle, and, if excessive, could significantly influence the trajectory.

Analysis

Assuming the rail is cantilevered from its base, the static deflections are given by the solution to

$$\mathbf{EI\ d^2y / dx^2 = F\ (L - x)},$$

where **E** = Young's modulus for the rail material,
I = Rail area moment of inertia,
y = Rail deflection,
x = Distance along the rail from its fixed end,
F = Force applied normal to the rail axis, and
L = Rail length.

Assuming the rocket starts from its base end, this can be readily integrated to give the slope at the free end (**x = L**):

$$\mathbf{dy / dx\ (L) = FL^2 / 2\ EI}.$$

This result will be multiplied by a factor of 2 to reflect dynamic overshoot.

The transverse force F has two components, that in pitch and that in yaw. The pitch component is

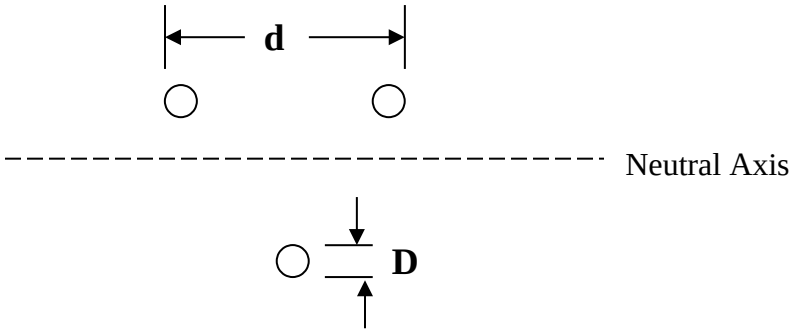
$$\mathbf{F = T\,\delta y + mg\cos(QE)},$$

where **T** = Thrust,
δy = Thrust misalignment pitch angle,
m = Rocket mass,
g = Acceleration due to gravity, and
QE = Quadrant Elevation angle.

The yaw component of the transverse force is just

$$\mathbf{F = T\,\delta z}.$$

A few comments about statistics are now in order. The gravity term in the pitch force is invariant. That is, it will always have the same value. But the two thrust misalignment forces are stochastic variables whose values will be different for every instantiation. Values for δ can be found from Ammons & Hoult[1]. When the two effects are compared for Gold Rush II, the lateral weight component = 55lb * cos(80°) = 9.55 lb, and the worst case (3 σ) thrust misalignment component = 400lb * 0.1° * 3 * π / 180 = 2.09 lb. Thus, the weight is by far the largest lateral force.



Moment of Inertia

Now suppose we use a ham radio tower for the rail structure as sketched above. These consist of three steel tubes tied together with a Z-shaped steel rods welded to the tubes. Typical dimensions are;

d = Face width = 12.5”
D = Tubing OD = 1.25” , and
t = Tubing wall thickness = 0.065”

[1] R. L. Ammons and C. P. Hoult, “Standardized Perturbation Values for Several Sounding Rockets to be Used in the Calculation of Dispersion and Structural Loads”, Aerojet-General Corporation Memo 8110:M0658:ak, 16 September 1970

The rod contributions to bending stiffness can be neglected. Now the neutral axis sketched above must be twice as far from the bottom tube as from the top two if it passes through the section area centroid. The area moment of inertia is

$$\mathbf{I = A\, d^2 / 2},$$

where **A** = cross section area of a tube.

A quick calculation shows that the yaw moment of inertia about a vertical neutral axis is the same. Since the cross section area of a tube is just

$$\mathbf{A = \pi\, D\, t}.$$

The bending stiffness of the radio tower is finally

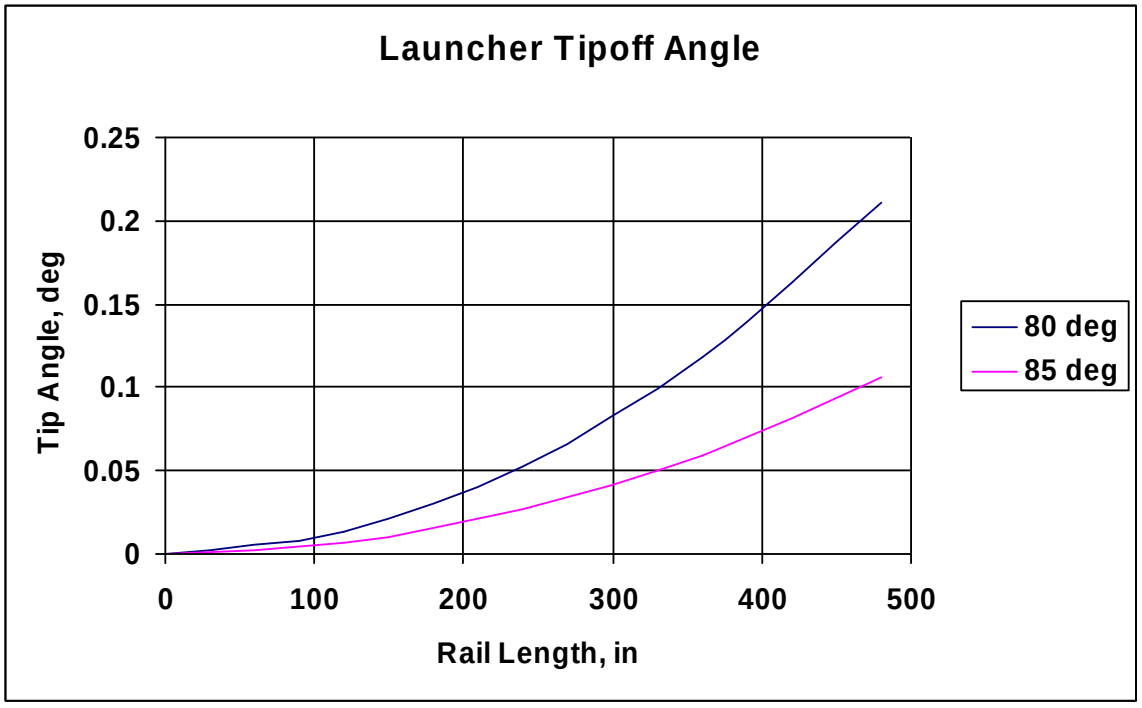
$$\mathbf{I = \pi\, D\, t\, d^2 / 2},$$

and the slope at the free end is

$$\mathbf{dy / dx\ (L) = FL^2 / \pi\, E\, D\, t\, d^2}.$$

Numerical Results

Some quick results are that the lateral weight component has an effect an order of magnitude larger than that from thrust misalignments as shown above. The plot below, for the Gold Rush II rocket, shows the pitch plane tipoff angle for two QE values:



Recommendations

Since ham radio towers come in 10 foot lengths, holding the rail length to 20' = 240" will ensure that the tipoff error will be less than about 0.1 deg., a good rule of thumb upper bound.

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